System Modeling And Agile Development Of An Online Bus Reservation System

Course Name: Introduction to Software Engineering

Course Code: IS1103-1

Assignment Number & Title: Assignment 2 – System Modeling and Agile Development of an Online Bus Reservation System

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Academic Year: 2025-2026

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Date of Submission: April 05, 2026

2. Abstract

This project allowed me to apply what I learned about creating a reservation system on the Web in a real-life application. However, the primary focus of the project was not just to create a reservation system, but to explore and analyze the entire process of creating a system. Another focus was to examine and learn about how systems are modeled and what methodologies are used in building systems prior to coding. More emphasis was given to modeling a system and developing that system in an organized manner than writing actual code.

I developed various UML models to visualize the system. A context diagram was proposed to represent how the system interacts with other parties like users or payment services. Understanding step by step booking process through sequence diagram.

I represented different states (searching, selecting booking & confirmation) using a state diagram.

I made different UML models to show how the system works. I used a context diagram to show how the system works with outside things like users, payment gateways, and bus operators. I made a sequence diagram to show how to book a ticket step by step. A state diagram showed and helps us to understand the different stages of a booking, such as searching, choosing correctly, correct payment without any issues, and final closing.[1]

For our daily work, I utilized a Scrum framework as a way to implement Agile methodology meaningfully and effectively. As a way to get tasks at hand completed, I divided tasks into incremental units of time (Sprints) that were customized to the team members to whom they would be assigned. I understood how the various members of the teams would earn their way to the finish line and what the finish line looked like.

Through this exercise, I learned how system development projects (i.e., software systems) are methodically planned/designed/built out from beginning to end. Overall this exercise provided me with a basis for understanding how the development process is broken down and improved upon rather than just diving right into coding, as mentioned above. [2]

3.Github Repository Link :

Github url: [Bhavyatha21/ISE-Task-2: System Modeling And Agile Development Of An Online Bus Reservation System]

4. Introduction

When developing software, it is important to have a design that can help in the successful implementation of a project. Unfortunately, most projects are started without proper structure and this affects project management. This case study was developed as an analysis into how the use of a system model can avoid similar mistakes in future projects.

This case study aimed to produce a UML system model representation of an online bus reservation system. UML diagrams are beneficial because they provide a visual representation of what the system will look like before actual code is produced [1]. As such, the user interaction (uses) with the system is easier to understand and the flow of data is also easier to understand.

The agile process applied to this task. The work was done using Scrum and separated into Sprints - a point of completion for a portion of the work to be accomplished. This is one of the most common ways to divide the work into small pieces for a project because of the continual improvement aspect related to agile development.[2]

As part of the System Modelling & Agile Methods unreflective unit for the 2020-2021 academic year, this assisted in understanding how the theory of System modelling and Agile methods is applied in real-life scenarios.

5. Problem Statement

By utilizing existing technology, the project aims to create an efficient, user-friendly ticket purchasing method that will provide an efficient transaction method to book a ticket for a bus in as little as 2 minutes.

The intended functionality of the system is as follows:

- Locate transportation options via bus from one location to a different location
- Choose your own seat
- Purchase tickets
- View your booking information
- Cancelling a ticket

The primary goal of this project is to develop a very clear and detailed system specification, and to have a strong Agile project plan.

There were some assumptions made about the system being built:

- Users will have internet access
- Payment processing will work as expected
- User’s data is accurate.

6. Dataset Description

This assignment does not pertain to analysing data; it is centred on designing systems. Therefore, dataset-related information is not applicable in this assignment.

7. Tools and technologies that were used

Programming Language: Not Applicable Tools

Used:

Draw.io (for creating UML diagrams)

GitHub (for tracking changes)

Microsoft Word (for creating documents)

8. Methodology / Architecture of the Model

8.1 Description of the Model

We employed multiple models for a better understanding of the system, each providing another perspective of how it operates.

Context model - It will explain how the system will relate to any external entity and details what the system does, as a basic overview, including how it is applied and those applications, but without going into significant detail, therefore assisting in determining the boundary definition of the systems [1].

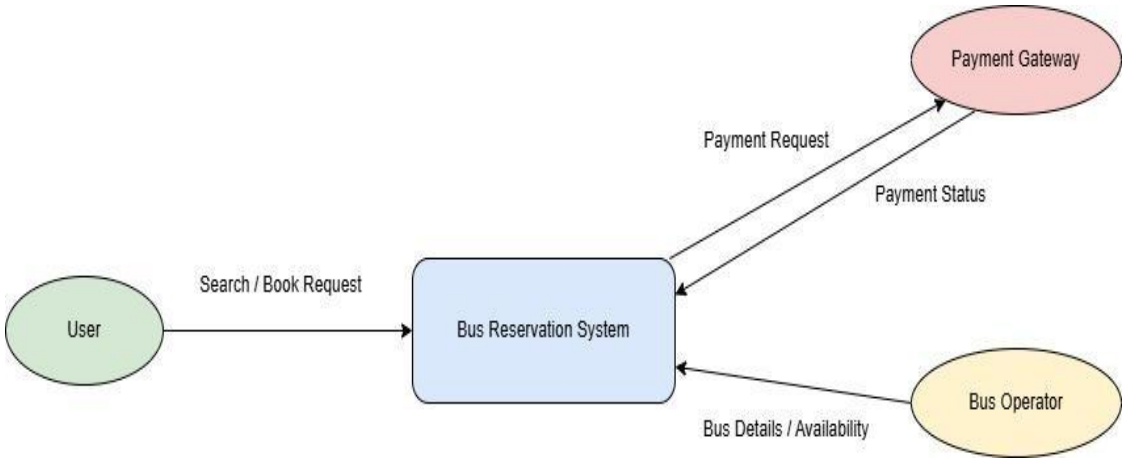
Sequence diagram - This diagram indicates the sequence of operations. It provides a stepby-step overview of how a user may interact/perform actions within the system, which is useful to assist in visualising the flow of actions when making a booking [3].

The different states of a booking are shown on the state diagram in a series of blocks to depict the process flow; for instance, searching for available seats, selecting an available seat, paying for the selected available seat, and confirming the booking. Additionally, the arrows between each of the five blocks represent the system's transition from one blocking to another [1].

All combined, these types of diagrams make it easy to visualize and understand how the system and user will interact with the system without having to actually see the underlying code.

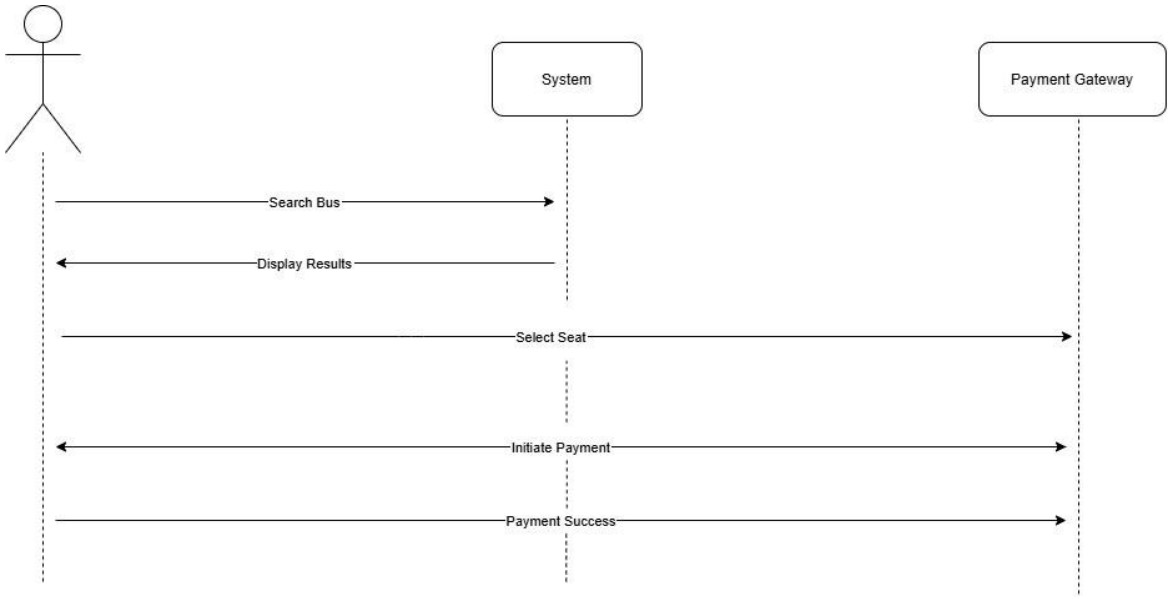
8.2 Diagrams

a) Context Diagram



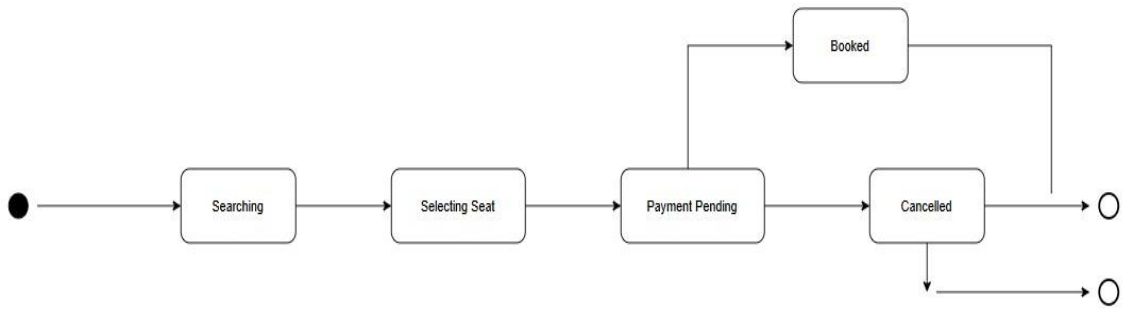
The state diagram also provides insight into how the user and the system interact; additionally, it provides a framework for analysing the interactions of the system with multiple external services or systems, providing a high-level view of the boundaries of the system.

b) Diagram of the Sequence (Booking Process)



This diagram shows how things happen when you book a ticket. It made it easy to see each step, from searching to confirming.

c) Diagram of the State (Booking lifecycle)



This diagram shows the different stages of a booking. It shows how the system goes from one step to the next.

9. Implementation Details

The project emphasized design, rather than coding. However, it was divided into logical units.

The principal components of the system are:

- User module (log-in; search)
- Scheduling module
- Payment module
- Administrative module

These units may be developed as separate components and integrated at a later date.

The system could also contain a multi-layered architecture consisting of a front-end layer, a back-end layer and a database. This type of architecture is frequently used in "realworld" systems.[2]

10. Agile Method (Scrum Approach)

Agile method was followed in a simple way. Not too detailed. Just basic idea.

Work was divided into small parts called sprints.

Example:

Sprint 1:

- Login
- Search bus

Sprint 2:

- Seat selection
- Booking

Sprint 3:

- Payment
- Confirmation User stories were also written.

Example:

- As a user, I want to search buses so that I can choose one
- As a user, I want to book ticket so that I can travel

Definition of done was simple. Task is complete when it works correctly.

Doing like this made work easier. Not confusing.

11. Results and Analysis

11.1 Quantitative Results

This assignment doesn't have any numbers because it doesn't involve data analysis or machine learning. The assessment hinges on the precision and lucidity of system models.

11.2 Graphs and Plots

Graphs and plots are not applicable. UML diagrams were used instead to represent the system.

11.3 Qualitative Results

Using the system modelling approach made it easier to understand the application in a clear way.

- The context diagram makes it easy to see how systems work together.
- The sequence diagram shows how to book step by step.
- The state diagram shows all the different states that a booking can be in.

The design of the system is user-friendly, simple, clear, and easy to understand and follow.

12. Discussion

Designing a system takes a lot of thought and it became clear and I realised it while I was working on this assignment. There are many steps and interactions in even a simple app.

At first, they didn't think about steps like confirming payment. These gaps were only visible after I drew diagrams. Agile methodology also helped me figure out how to break big tasks down into smaller ones.

When conducting this assignment, I noticed that even the most straightforward systems have many internal steps. Initially, several of those steps were missed, such as the completion of payment confirmation, but using diagrams helped to visualize those missing pieces.

Agile methodology was helpful in breaking up the work into smaller tasks. This approach allowed for improved management of the total number of tasks rather than attempting to complete the entire list of tasks at once.

13. Regularization and Optimization

Not relevant for this task.

14. Limitations and Future Work

The system created is very simple and does not have many advanced features such as:

- Updates on seat availability in real time
- Pricing that changes
- Systems for making suggestions

These features can be developed in the future as enhancements to the system, and additional security features can also be included as future enhancements to the system.

15. Conclusion

By doing this assignment, I gained insight into both the use of system modelling as well as Agile Development methodologies.

The use of UML diagrams helped me visualize the relationship between the various components of the system, while the use of Scrum as an organisational method for software development allowed me to create an effective means of completing the development process.

Overall, the assignment provided me with a deeper understanding of concepts relating to Software Engineering.

In summary, the completion of this assignment provided me with a clear understanding of how actual systems can be built and modified over time [1].

16. References

[1] I. Sommerville, *Software Engineering*, 10th Edition, Pearson, 2016.
[2] Agile Alliance, "Agile Methodology Guide."
[3] Draw.io Documentation

17. Appendix

Here is where you can add any diagrams, draft designs, or pertinent notes related to system modelling can be included here if required. I have included all necessary aspects and all pertinent information within the main sections of this assignment.

18. Declaration of Academic Integrity

I declare that this assignment is my own work, and I have not copied or plagiarized anything from other sources. I have cited all references and resources used appropriately, and any ambiguity may result in my forfeiting of marks

Student Signature

Date: April 05, 2026